EVM Dashboard — User Manual

CE 725 Construction Economics and Finance · Course project, Topic 7

1. Purpose and intended users

The EVM Dashboard tells a project team, at each monthly (or weekly) status date, whether a construction project is earning the value it planned to and at what cost, and what the final cost is likely to be under clearly stated assumptions. It is intended for project managers, construction managers, project-controls engineers, contractors, clients/owners and quantity surveyors. It supports judgement; it does not replace it.

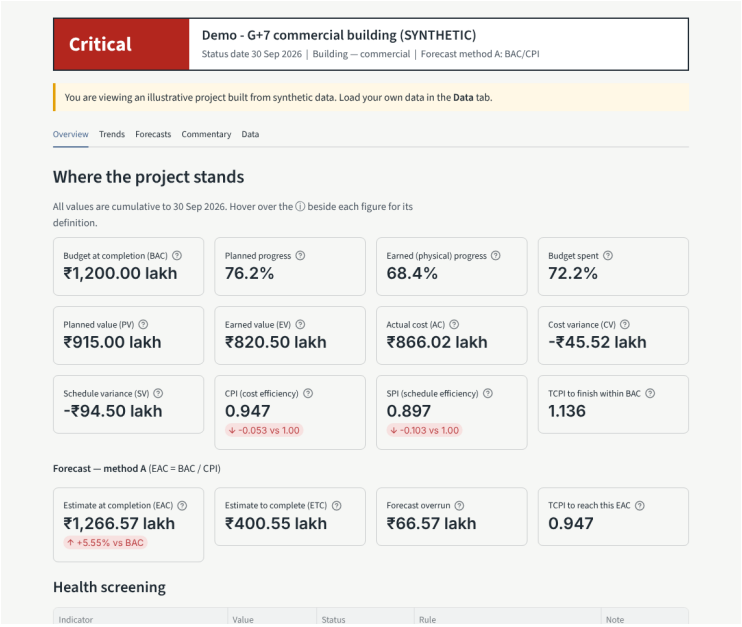
2. System requirements and launch

- Python 3.10 or later (tested on 3.13), and a modern browser.
- Install once, then launch:

```
cd evm_dashboard
python -m pip install -r requirements.txt
streamlit run app.py
```

The dashboard opens at `http://localhost:8501` showing a **synthetic demonstration project** (G+7 commercial building, 6 work packages, 9 monthly reports). No internet connection is needed for any calculation.

3. Screen layout



Overview screen: status signboard, sidebar controls and KPI tiles

- **Signboard** (top): overall health label, project, status date and forecast method.

- **Sidebar**: forecast method (A–D), money display unit (full, lakh, crore, million — display only), reporting date to view, and health thresholds.
- **Tabs**: Overview, Trends, Forecasts, Commentary, Data.

4. Setting up a project — the Data tab

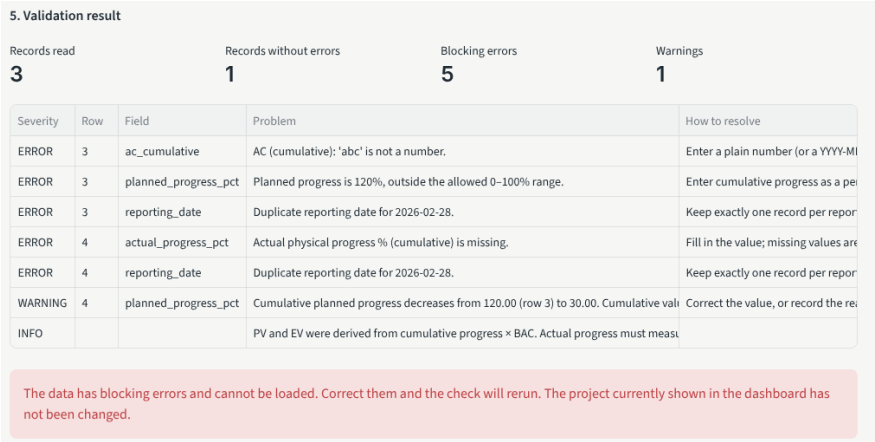
Step 1 — choose the kind of data you have.

Mode	You supply	The tool derives
Progress-Based Project Data	BAC, cumulative planned %, cumulative actual **physical** %, AC	PV = planned % × BAC; EV = actual % × BAC
Pre-calculated EVM Data	cumulative PV, EV, AC (e.g. exported from scheduling software)	nothing — supplied values are used as-is
Work-Package-Level Data	budget, planned %, actual %, cumulative AC per work package per date	budget-weighted PV and EV, summed AC

Download the **template** for your mode (or a **sample dataset**) from the buttons under Step 1. Blank templates are also in `templates/`.

Step 2 — provide the data.

Upload a file (CSV or .xlsx; choose the worksheet for Excel). The tool shows a preview, suggests a **column mapping** from your headers (check it — unmapped columns are ignored, never guessed), asks for **project settings** (name, type, currency, BAC if not in the file, baseline dates, reporting frequency, date format), then shows the **validation result**: records read, records without errors, blocking errors, warnings, and a table giving the row, field, problem and how to fix it. If there are blocking errors there is no load button and the project already shown is not changed. If validation passes, tick *Replace the project currently shown* and press **Load into dashboard**.



Validation of a faulty upload: five blocking errors with row numbers; the load button is withheld

Enter or edit manually — an editable table with only the columns your mode needs (* = required). Add rows with the **+** control. *Reload from current project* edits the loaded data; *Start a new blank table* (tick the confirmation first) clears it. The same validation rules as uploads apply, and nothing reaches the dashboard until you load it.

Load a built-in example — the demonstration project or the two-period benchmark case used for validation.

Exports (top of the Data tab): Results (CSV), Inputs (CSV) and a Workbook (Excel) containing project settings, validated inputs, reporting periods, all results with status notes, work-package detail and validation messages.

5. Input rules that matter

- **Cumulative means to-date totals.** Never add cumulative values; if you only have monthly costs, use the `ac_period` column and the tool will total them.
- **Actual progress is physical**: the share of budgeted scope completed (quantities installed, milestones achieved). Time elapsed and MS Project's "% Complete" (a duration measure) are not physical progress.
- **Blank is missing, not zero.** A missing required value blocks loading. An AC of genuinely zero is allowed but makes CPI undefined.
- **Percentages are 0–100.** 0.45 means 0.45%, not 45% (the tool warns if all values look like fractions).
- **Work-package data:** every work package at every date; budgets constant; use a zero-budget package for unallocated costs.

6. Reading the results

Measure	Meaning	Good sign
PV / EV / AC	Budgeted value scheduled / budgeted value earned / cost actually spent (cumulative)	—
CV = EV − AC	Cost gain (+) or loss (−) on the work done	≥ 0
SV = EV − PV	Value of work ahead (+) or behind (−) plan — **money, not days**	≥ 0
CPI = EV/AC	Value earned per rupee spent	≥ 1.00
SPI = EV/PV	Rate of earning value versus plan	≥ 1.00
EAC, ETC, VAC	Forecast final cost, cost still to spend, budget minus forecast	VAC ≥ 0
TCPI to BAC	Efficiency the remaining work needs to finish within BAC	close to CPI

Undefined means the value cannot be calculated (e.g. CPI when AC = 0) and the reason is in the ⓘ tooltip; **N/A** means the measure does not apply (e.g. TCPI once all work is earned); ⚠️ means a value is shown but should be reviewed. None of these is ever shown as zero.

Health signboard. Defaults: CPI or SPI below 1.00 = Warning, below 0.90 = Critical; forecast overrun above 0% = Warning, above 10% = Critical. The overall label is Critical if any indicator is critical, *Insufficient data* if any indicator cannot be calculated, otherwise the worst of the rest. These are adjustable screening rules, not industry standards.

7. Forecasting methods (Forecasts tab)

Method	Formula	Use when
A	EAC = BAC / CPI	past cost efficiency is expected to continue
B	EAC = AC + (BAC − EV)/(CPI × SPI)	recovering the schedule is expected to cost extra (usually the most pessimistic)
C	EAC = AC + (BAC − EV)	the past variance was a one-off and its cause has been removed (optimistic)
D	EAC = AC + your ETC	the team has re-estimated the remaining work bottom-up

The Forecasts tab compares all four side by side, charts BAC against each EAC and shows how EAC has moved over time. For method D, enter the independent ETC for the selected date and press **Apply estimate**; it is labelled user-entered in exports and lasts for the session.



Trends tab: cumulative PV/EV/AC with BAC and forecast EAC; CPI and SPI trends

8. Charts (Trends tab)

Cumulative PV/EV/AC with BAC and the forecast EAC (dashed); CPI and SPI trends with the 1.00 reference and critical line; planned vs actual progress; budget vs forecast final cost; and, for work-package data, CV and SV by work package. Solid lines are observed data, dashed lines are forecasts, and a missing reporting period appears as a break rather than a joined line.

9. Commentary tab

Plain-English text generated by fixed rules from the numbers: observed facts, change since the previous report, the forecast and its assumption, suggested investigations, and caveats. It never states a cause the data cannot show (it will say cost efficiency is unfavourable, not that labour productivity fell). Notes you entered for the period are quoted and marked as unverified. Download it as Markdown for a monthly report.

10. Common messages and troubleshooting

Message	What to do
Required column '…' is not mapped	Map it in Step 3 or add the column from the template.
'…' is not a number	Remove text from the numeric cell; plain numbers, separators and ₹ are fine.
outside the allowed 0–100% range	Progress must be a percentage of scope between 0 and 100.
Duplicate reporting date	Keep one row per date (per work package in WP mode).
Work package(s) … have no record for …	Repeat the unchanged cumulative values for that WP and date.

Message	What to do
Cumulative … decreases (warning)	Correct it, or explain it (re-measurement, credit note) in the notes column.
BAC in the file differs from settings	Make them equal, or clear the settings BAC.
Missing monthly period (warning)	Add the period or set the reporting frequency to Irregular.
App will not start	Check `python -m pip install -r requirements.txt` completed; run from the `evm_dashboard` folder.
Edits disappeared	Session state is not saved; export the workbook before closing.

11. Assumptions and limitations (summary)

Forecasts hold only under their stated assumptions. SPI is a budget-value measure and does not show critical-path delay. One baseline and one currency per dataset. Rounded percentage inputs lose precision. All bundled data is synthetic. The full list is in `documentation/assumptions_and_limitations.md`.